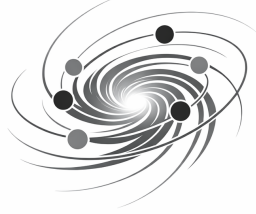


Part of the Monistic Continuum Model (MCM)



Electron Entanglement (EE)

Spectral and Topological Structures

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$$|\Psi_{12}\rangle = \hat{P}_{\text{QM}}(\mathcal{K}_{12}), \quad \mathcal{K}_{12} = \bigcup_{i=1}^N c_i, \quad c_i \subseteq W_n.$$

The quantum-mechanical two-electron configuration arises as a spectral projection of the underlying coupling geometry. The structure $\mathcal{K}_{12}$ is the union of all continuous sub-geometries c_i , each of which emerges from the vortex modes W_n .

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1 Abstract

This work investigates how electron entanglement can arise as a continuum-geometric phenomenon within the Monistic Continuum Model (MCM). Electron entanglement is traditionally formulated

within the linear Hilbert-space framework of quantum mechanics, where nonlocal correlations emerge from tensor-product structures and unitary evolution. In contrast, the Monistic Continuum Model (MCM) describes electrons as stable, vortex-like excitations of a continuous substrate $C(x, \lambda)$ defined on a differentiable manifold M . Within this framework, entanglement arises not from abstract state-space composition but from geometric and dynamical properties of the continuum itself.

We introduce the continuum field

$$C : M \times \mathbb{R} \rightarrow V,$$

its correlation kernel

$$K(x, y; \lambda) = \langle C(x, \lambda), C(y, \lambda) \rangle_V,$$

and the dynamical operator

$$D = D_{\text{loc}} + D_{\text{nl}},$$

which governs both local vortex stability and nonlocal coherence. Electron modes are defined as localized, topologically protected vortex solutions M_i of the continuum dynamics, characterized by invariants such as phase winding, circulation, and geometric mass functionals.

Entanglement emerges when two such modes M_1 and M_2 share a non-separable continuum region encoded by the nonlocal term D_{nl} , producing a joint configuration

$$C = M_1 + M_2 + A_C,$$

where A_C denotes the continuum-mediated coherence field.

Coarse-graining over the evolution parameter λ induces an effective operator whose spectral decomposition yields linearity, superposition, and tensor-product composition. Under assumptions of kernel regularity, mode stability, and coherence persistence, we show that nonlocal continuum coherence generates a mathematically non-separable two-mode configuration equivalent to quantum entanglement. This establishes entanglement as a continuum-geometric effect rather than a purely algebraic construct.

Finally, we derive experimentally accessible predictions, including geometry-dependent correlation plateaus, kernel-induced coherence modulation, and deviations from exchange symmetry. These signatures provide a pathway for distinguishing MCM-based entanglement from standard quantum mechanics and offer a continuum-theoretic foundation for nonlocal quantum phenomena.

2 Introduction

This section addresses the question: Why is a continuum-based reformulation of electron entanglement necessary? Electron entanglement is conventionally formulated within the linear Hilbert-space

framework of quantum mechanics, where composite systems are represented by tensor-product structures and nonlocal correlations arise from the algebraic properties of state vectors. While this formulation is operationally successful, it leaves several conceptual and structural questions unresolved: the ontological status of the wavefunction, the origin of nonlocal correlations, the physical meaning of tensor-product composition, and the absence of a geometric mechanism underlying entanglement. These issues motivate the search for a deeper, continuum-based reformulation. In the Monistic Continuum Model (MCM), electrons are not point-like particles but stable, vortex-like excitations of a continuous substrate

$$C : M \times \mathbb{R} \rightarrow V,$$

where M is a differentiable manifold, V an internal state space, and λ an evolution parameter. The continuum possesses a correlation kernel

$$K(x, y; \lambda) = \langle C(x, \lambda), C(y, \lambda) \rangle_V,$$

which encodes both local and nonlocal structure. The dynamics of the continuum are governed by an operator

$$D = D_{\text{loc}} + D_{\text{nl}},$$

where D_{loc} stabilizes vortex modes and D_{nl} generates nonlocal coherence. This decomposition provides a geometric and dynamical foundation for phenomena that, in standard quantum mechanics, appear only as algebraic constructs.

A central motivation for the continuum approach is the need for a physically grounded mechanism for entanglement. In the MCM, entanglement arises when two electron modes M_1 and M_2 share a non-separable continuum region, producing a joint configuration

$$C = M_1 + M_2 + A_C,$$

where A_C is a coherence field induced by D_{nl} . This formulation replaces the abstract tensor-product structure with a geometric non-separability condition,

$$K(x_1, x_2; \lambda) \neq K(x_1; \lambda) K(x_2; \lambda),$$

which directly expresses continuum-level correlation.

The continuum perspective also provides a natural route to the emergence of Hilbert-space structures. Coarse-graining over λ yields an effective linear operator

$$\mathcal{H} = \int D w(\lambda) d\lambda,$$

whose spectral decomposition induces superposition, linearity, and tensor-product composition. Thus, quantum mechanics appears as an emergent theory derived from deeper continuum dynamics. Comparisons with quantum field theory (QFT) highlight further motivations. QFT treats fields as operator-valued distributions but does not provide a geometric mechanism for nonlocal correlations. In contrast, the MCM attributes nonlocality to the structure of the correlation kernel K and the operator D_{nl} , offering a physically interpretable origin for entanglement. Similarly, general relativity (GR) describes spacetime geometry but does not incorporate nonlocal quantum correlations; the MCM suggests a unified geometric substrate capable of supporting both local curvature and nonlocal coherence.

The continuum-based reformulation is therefore necessary to address foundational gaps in standard quantum theory: the origin of nonlocality, the physical meaning of entanglement, the emergence of Hilbert-space structures, and the geometric interpretation of quantum coherence. By grounding entanglement in continuum geometry and dynamics, the MCM provides a coherent and physically motivated framework that extends and deepens the conceptual foundations of quantum mechanics.

3 Related Work

This section investigates the question: How does the MCM relate to existing nonlocal, geometric, and relational quantum frameworks? The Monistic Continuum Model (MCM) proposes that

electrons are stable, vortex-like excitations of a continuous substrate

$$C : M \times \mathbb{R} \rightarrow V,$$

with dynamics governed by an operator

$$D = D_{\text{loc}} + D_{\text{nl}},$$

and correlations encoded in a kernel

$$K(x, y; \lambda) = \langle C(x, \lambda), C(y, \lambda) \rangle_V.$$

To situate this framework within existing approaches to nonlocality and geometry in quantum theory, we compare it with standard quantum mechanics (QM), quantum field theory (QFT), geometric and topological formulations, and relational quantum frameworks.

Ontological levels of existing frameworks

A structural distinction clarifies how the MCM relates to other approaches. Standard QM and QFT operate on an *algebraic ontology*: states are elements of Hilbert spaces, and correlations arise from operator algebra. Geometric and topological approaches introduce manifold structure but retain Hilbert-space primacy. Relational frameworks reinterpret correlations but do not provide a physical substrate. GR is geometric but non-quantum. The MCM differs by combining a *geometric ontology* with *nonlocal operator dynamics*, positioning it uniquely among these frameworks.

Standard quantum mechanics (QM)

In conventional QM, entanglement arises from the tensor-product structure of Hilbert spaces and the linearity of unitary evolution. Composite systems are represented by states $\psi \in \mathcal{H}_1 \otimes \mathcal{H}_2$, and nonlocal correlations are encoded in the algebraic non-separability of ψ . However, QM does not provide a geometric mechanism for nonlocality; the wavefunction resides in an abstract state space rather than on a physical continuum. In contrast, the MCM grounds nonlocality in the structure of the correlation kernel K and the nonlocal component D_{nl} , with entanglement emerging from non-separable continuum configurations rather than purely algebraic tensor products.

Quantum field theory (QFT)

Quantum field theory extends QM by promoting fields to operator-valued distributions on spacetime, with locality enforced through commutation relations and microcausality. While QFT successfully describes particle creation, annihilation, and interactions, its treatment of entanglement remains tied to Hilbert-space factorization and region-based algebras. The MCM differs by treating the continuum $C(x, \lambda)$ as a physically real substrate whose geometric and topological properties

directly generate electron modes and their correlations. Nonlocality is encoded in the kernel K and the operator D_{nl} , rather than in the algebra of local field operators alone.

Geometric and topological quantum frameworks

Geometric and topological approaches to quantum theory, including fiber-bundle formulations, geometric quantization, and topological quantum field theories, introduce manifold structures, connections, and invariants to describe quantum states and phases. These frameworks emphasize holonomies, Berry phases, and topological protection but typically retain the Hilbert-space ontology. The MCM shares the geometric emphasis—electron modes are vortex-like structures with topological invariants—yet it goes further by treating the continuum as ontologically primary and Hilbert-space structures as emergent from coarse-grained dynamics and spectral properties of D .

Relational and nonlocal frameworks

Relational and nonlocal frameworks, such as relational quantum mechanics, consistent histories, and certain nonlocal hidden-variable models, focus on correlations and relations rather than absolute states. They reinterpret the meaning of the wavefunction and measurement but often lack a concrete geometric substrate that carries these relations. In the MCM, relations between modes M_1 and M_2 are realized as shared continuum regions and non-separable kernel structures:

$$K(x_1, x_2; \lambda) \neq K(x_1; \lambda) K(x_2; \lambda),$$

providing a physically interpretable basis for relational properties.

General relativity (GR)

From the perspective of general relativity, spacetime geometry is central, but quantum nonlocality is not incorporated into the classical manifold structure. The MCM suggests a unified geometric substrate in which both local curvature and nonlocal coherence can, in principle, be encoded in the continuum field C and its kernel K . This positions the MCM as a candidate framework for bridging geometric and quantum aspects of reality.

Summary

The MCM is related to existing nonlocal, geometric, and relational quantum frameworks by sharing their emphasis on geometry, topology, and correlation, while differing in its ontological commitment to a continuous substrate and its derivation of entanglement from non-separable

continuum configurations. This comparison motivates the subsequent formal development of the MCM and clarifies its role as a continuum-based extension of standard quantum theory.

4 Mathematical Foundations of the MCM

This section addresses the question: What is the precise mathematical structure of the continuum substrate $C(x, \lambda)$? In the Monistic Continuum Model (MCM), the fundamental physical entity is

a continuous substrate rather than discrete particles. This substrate is represented by a field

$$C : M \times \Lambda \rightarrow V,$$

where M is a smooth, connected, oriented n -dimensional manifold (typically $n = 3$ for physical space or $n = 4$ for spacetime), $\Lambda \subseteq \mathbb{R}$ is an evolution-parameter domain, and V is a finite-dimensional internal state space equipped with an inner product $\langle \cdot, \cdot \rangle_V$. The pair (M, V) defines the geometric and internal degrees of freedom of the continuum.

Continuum configuration space and function spaces

The configuration space of the continuum is defined as

$$\mathcal{C} = \{C \mid C \in \mathcal{F}(M \times \Lambda, V)\},$$

where $\mathcal{F}(M \times \Lambda, V)$ denotes an appropriate function space, such as $C^\infty(M \times \Lambda, V)$ or a Sobolev space $H^k(M \times \Lambda, V)$, depending on regularity requirements. For fixed $\lambda \in \Lambda$, the instantaneous configuration is

$$C_\lambda : M \rightarrow V, \quad C_\lambda(x) := C(x, \lambda),$$

and the space of instantaneous configurations is

$$\mathcal{C}_\lambda = \{C_\lambda \mid C_\lambda \in \mathcal{F}(M, V)\}.$$

We equip $\mathcal{C}_\lambda$ with an inner product

$$\langle C_\lambda, C'_\lambda \rangle_{\mathcal{C}} = \int_M \langle C_\lambda(x), C'_\lambda(x) \rangle_V d\mu(x),$$

where $d\mu$ is a measure induced by a metric g on M . Under suitable completeness assumptions, $\mathcal{C}_\lambda$ becomes a Hilbert space, providing a natural bridge to standard quantum structures.

Correlation kernel and nonlocal structure

The continuum substrate possesses a correlation kernel

$$K : M \times M \times \Lambda \rightarrow \mathbb{C}, \quad K(x, y; \lambda) = \langle C(x, \lambda), C(y, \lambda) \rangle_V,$$

which encodes both local and nonlocal correlations. For each fixed λ , the kernel $K_\lambda(x, y)$ defines an integral operator

$$(\mathcal{K}_\lambda f)(x) = \int_M K_\lambda(x, y) f(y) d\mu(y),$$

acting on suitable function spaces over M .

The kernel is *separable* if it factorizes as

$$K_\lambda(x, y) = k_\lambda(x) k_\lambda(y),$$

and *non-separable* otherwise. Non-separability of K_λ expresses continuum-level entanglement: correlations cannot be reduced to products of single-point contributions.

Dynamical operator and decomposition

The evolution of the continuum is governed by a dynamical operator

$$D = D_{\text{loc}} + D_{\text{nl}},$$

acting on $\mathcal{C}$. The local part D_{loc} is typically a differential operator constructed from the metric g and possibly additional structures such as connections or potentials, e.g.

$$D_{\text{loc}} = a_1 \Delta_g + a_2 \nabla_g + a_3,$$

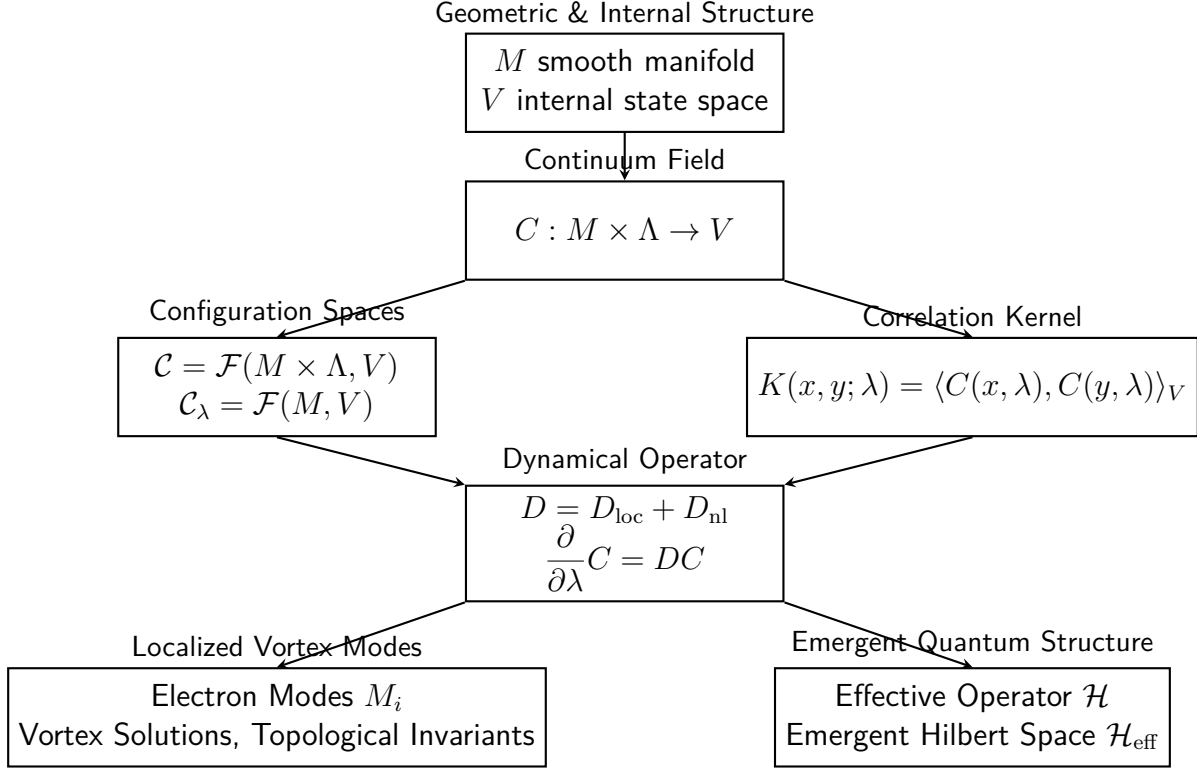
where Δ_g is the Laplace–Beltrami operator and ∇_g a covariant derivative. The nonlocal part D_{nl} is defined via a kernel $\mathcal{J}$,

$$(D_{\text{nl}} C)(x, \lambda) = \int_M \mathcal{J}(x, y; \lambda) C(y, \lambda) d\mu(y),$$

with $\mathcal{J}$ related to K . The full dynamics are given by the evolution equation

$$\frac{\partial}{\partial \lambda} C(x, \lambda) = D C(x, \lambda),$$

which generalizes local field equations by incorporating nonlocal coherence.



Electron modes as vortex solutions

Electron modes are defined as stable, localized solutions $M_i(x, \lambda)$ of the continuum dynamics, satisfying

$$DM_i = 0,$$

or more generally

$$\frac{\partial}{\partial \lambda} M_i = DM_i,$$

with appropriate stability conditions. These modes possess topological and geometric invariants—such as phase winding numbers, circulation integrals, and energy functionals—which correspond to physical quantities like spin, charge, and mass. Mathematically, they can be modeled as vortex-like structures in C_λ , for example as zeros of C_λ with nontrivial phase winding in V .

Emergent Hilbert-space structure

Under coarse-graining over λ , we define an effective operator

$$\mathcal{H} = \int_{\Lambda} w(\lambda) D_\lambda d\lambda,$$

where $w(\lambda)$ is a weight function and D_λ the restriction of D at parameter λ . Spectral decomposition of $\mathcal{H}$ yields eigenmodes

$$\mathcal{H}\phi_n = E_n\phi_n,$$

which form a basis of an emergent Hilbert space $\mathcal{H}_{\text{eff}}$. Superposition, linearity, and tensor-product composition arise as effective structures derived from continuum dynamics rather than as fundamental postulates.

Comparison with QM, QFT, and GR

In standard QM, the primary structure is a Hilbert space $\mathcal{H}$ of state vectors, with dynamics governed by a Hamiltonian operator H . The MCM, by contrast, begins with a continuum field C on a manifold M and derives an effective Hilbert space from its dynamics. In QFT, fields are operator-valued distributions on spacetime, but nonlocal correlations are encoded in algebraic structures rather than in a correlation kernel of a physically real continuum. The MCM introduces a concrete kernel K and a nonlocal operator D_{nl} as generators of coherence. GR provides a geometric manifold with metric g , but does not incorporate quantum nonlocality; the MCM extends geometric structure by adding an internal state space V and a continuum field C capable of supporting both local curvature and nonlocal coherence.

Result: precise mathematical structure of the continuum

The continuum substrate $C(x, \lambda)$ is thus specified as a field on a smooth manifold M with values in an internal state space V , equipped with a correlation kernel K , a dynamical operator $D = D_{\text{loc}} + D_{\text{nl}}$, and a function-space structure that admits an emergent Hilbert space. This mathematical formulation provides the foundation for treating electron entanglement as a continuum-geometric phenomenon and for deriving quantum-like structures from deeper continuum dynamics.

5 Electron Modes as Vortex Solutions

This section investigates the question: How do stable vortex modes arise as electron configurations in the continuum? In the Monistic Continuum Model (MCM), electrons are modeled not as

point-like particles but as stable, localized vortex solutions of the continuum field

$$C : M \times \Lambda \rightarrow V,$$

where M is a smooth manifold, $\Lambda \subseteq \mathbb{R}$ an evolution-parameter domain, and V an internal state space. The goal of this section is to provide a precise mathematical characterization of such vortex modes, to identify their invariants, and to show how they correspond to electron configurations.

Vortex configurations in the continuum

For fixed $\lambda \in \Lambda$, we consider the instantaneous field

$$C_\lambda : M \rightarrow V, \quad C_\lambda(x) := C(x, \lambda).$$

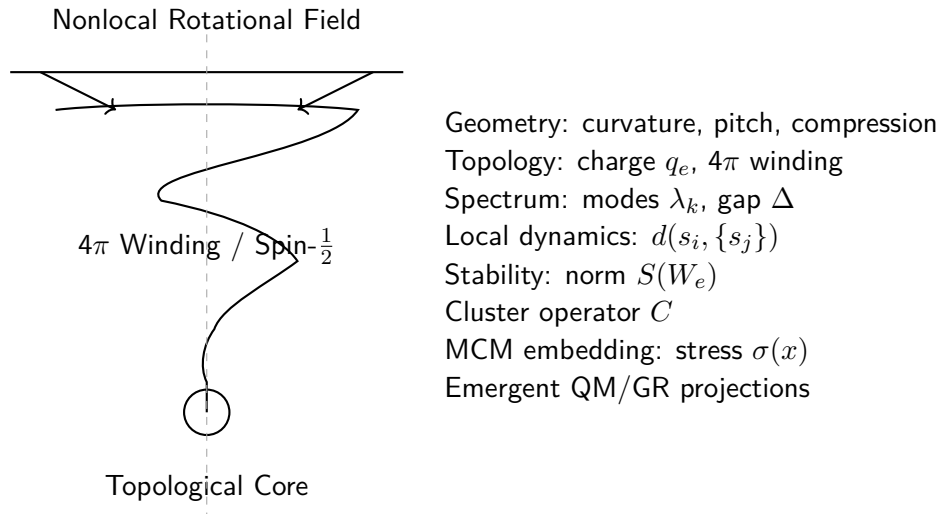
A vortex configuration is defined as a localized region in M where C_λ exhibits nontrivial phase structure or circulation. Let γ be a closed loop in M encircling a region where C_λ vanishes or is singular in an appropriate sense. We define the phase map

$$\theta_\lambda : M \setminus Z_\lambda \rightarrow S^1,$$

where Z_λ is the set of zeros or singularities of C_λ , and the winding number

$$w(\gamma; \lambda) = \frac{1}{2\pi} \oint_\gamma d\theta_\lambda.$$

A nonzero integer $w(\gamma; \lambda)$ indicates a vortex-like structure. Electron modes are identified with vortex configurations that are dynamically stable under the evolution generated by the operator D .



Dynamical equations and stability

The continuum dynamics are governed by

$$\frac{\partial}{\partial \lambda} C(x, \lambda) = D C(x, \lambda),$$

with

$$D = D_{\text{loc}} + D_{\text{nl}}.$$

We seek solutions $M_i(x, \lambda)$ that are localized, vortex-like, and stable. A stationary vortex mode satisfies

$$\frac{\partial}{\partial \lambda} M_i(x, \lambda) = 0,$$

which implies

$$D M_i = 0.$$

More generally, a dynamically stable mode satisfies

$$\frac{\partial}{\partial \lambda} M_i = D M_i,$$

with stability conditions such as bounded energy and preserved topological invariants. We define an energy functional

$$E[M_i] = \int_M \mathcal{E}(M_i(x, \lambda), \nabla M_i(x, \lambda)) d\mu(x),$$

where $\mathcal{E}$ is a local energy density depending on C and its derivatives. Stability requires that $E[M_i]$ is minimized (or at least stationary) under variations of M_i within an appropriate function space, subject to topological constraints.

Topological and geometric invariants

Electron modes as vortex solutions possess invariants that correspond to physical quantities. The winding number w is associated with spin-like properties, while circulation integrals

$$\Gamma(\gamma; \lambda) = \oint_{\gamma} \mathbf{j}_{\lambda} \cdot d\ell,$$

where $\mathbf{j}_{\lambda}$ is a current derived from C_{λ} , can be related to charge-like quantities. Mass-like properties can be associated with energy functionals and localization scales. These invariants are robust

under continuous deformations of the field that do not cross singularities, providing topological protection for electron modes.

Function spaces and mode classification

We consider the space of admissible vortex configurations

$$\mathcal{M}_{\text{vortex}} = \{M_i \mid M_i \in \mathcal{F}(M \times \Lambda, V), DM_i = 0, \text{vortex conditions}\},$$

where vortex conditions include nontrivial winding numbers, localization, and finite energy. This space can be decomposed into sectors labeled by topological invariants, e.g.

$$\mathcal{M}_{\text{vortex}} = \bigoplus_{w \in \mathbb{Z}} \mathcal{M}_w,$$

where $\mathcal{M}_w$ contains modes with winding number w . Electron configurations correspond to specific sectors, such as $w = \pm 1$, with additional constraints on energy and localization.

Two-mode configurations and interaction

When two electron modes M_1 and M_2 are present, the total field can be written as

$$C(x, \lambda) = M_1(x, \lambda) + M_2(x, \lambda) + A_C(x, \lambda),$$

where A_C is a coherence field induced by the nonlocal operator D_{nl} . The interaction between modes is mediated by the continuum and encoded in the correlation kernel

$$K(x, y; \lambda) = \langle C(x, \lambda), C(y, \lambda) \rangle_V.$$

Non-separability of K in regions where both modes are present indicates entanglement-like correlations at the continuum level.

Comparison with QM, QFT, and GR

In standard QM, electrons are described by wavefunctions $\psi(x)$ in a Hilbert space, with no explicit vortex structure in physical space unless additional constructs such as phase singularities are introduced. In QFT, electrons are excitations of a fermionic field, but the field is operator-valued and the notion of a classical vortex is not fundamental. The MCM, by contrast, treats electron configurations as vortex solutions of a physically real continuum field, with topological invariants corresponding to physical properties. GR provides a geometric manifold but does not describe

matter as vortex modes of a continuum; the MCM adds an internal field structure capable of supporting such configurations.

Result: electron modes as stable vortex solutions

Electron configurations in the MCM can be modeled as stable vortex solutions of the continuum field $C(x, \lambda)$, characterized by topological and geometric invariants, governed by a dynamical operator D , and classified within appropriate function spaces. These vortex modes provide a concrete, continuum-based realization of electrons, forming the basis for subsequent analysis of entanglement as a continuum-geometric phenomenon.

6 Emergence of Quantum Mechanics

This section addresses the question: How do Hilbert-space structures and linear quantum mechanics emerge from continuum dynamics? In the Monistic Continuum Model (MCM), the fundamental

entity is the continuum field

$$C : M \times \Lambda \rightarrow V,$$

with dynamics governed by

$$\frac{\partial}{\partial \lambda} C(x, \lambda) = D C(x, \lambda), \quad D = D_{\text{loc}} + D_{\text{nl}}.$$

Quantum mechanics, in contrast, begins with a Hilbert space $\mathcal{H}$ of state vectors and a Hamiltonian operator H . The aim of this section is to show how Hilbert-space structures and linear quantum dynamics can emerge from the deeper continuum dynamics of C .

Instantaneous configuration space and inner product

For fixed $\lambda \in \Lambda$, the instantaneous continuum configuration is

$$C_\lambda : M \rightarrow V, \quad C_\lambda(x) := C(x, \lambda).$$

We define the space of admissible configurations

$$\mathcal{C}_\lambda = \{C_\lambda \mid C_\lambda \in \mathcal{F}(M, V)\},$$

equipped with the inner product

$$\langle C_\lambda, C'_\lambda \rangle_{\mathcal{C}} = \int_M \langle C_\lambda(x), C'_\lambda(x) \rangle_V d\mu(x),$$

where $d\mu$ is induced by a metric g on M . Under suitable completeness conditions, $\mathcal{C}_\lambda$ becomes a Hilbert space. This provides a first step toward an emergent quantum structure: continuum configurations at fixed λ form a Hilbert space of states.

Correlation kernel and effective operators

The correlation kernel

$$K(x, y; \lambda) = \langle C(x, \lambda), C(y, \lambda) \rangle_V$$

defines an integral operator

$$(\mathcal{K}_\lambda f)(x) = \int_M K(x, y; \lambda) f(y) d\mu(y),$$

acting on functions $f : M \rightarrow \mathbb{C}$. Under appropriate conditions (symmetry, positivity, compactness), $\mathcal{K}_\lambda$ admits a spectral decomposition

$$\mathcal{K}_\lambda \phi_n(x; \lambda) = \kappa_n(\lambda) \phi_n(x; \lambda),$$

with eigenfunctions ϕ_n forming an orthonormal basis of a Hilbert space $\mathcal{H}_\lambda$. These eigenfunctions can be interpreted as emergent quantum modes, and the kernel $\mathcal{K}_\lambda$ as an effective correlation operator.

Coarse-graining over the evolution parameter

To obtain a time-like evolution analogous to quantum dynamics, we introduce a coarse-graining over λ . Let $w(\lambda)$ be a weight function defining an effective averaging, and define

$$\mathcal{H}_{\text{eff}} = \int_\Lambda w(\lambda) \mathcal{H}_\lambda d\lambda,$$

in the sense of a direct integral or suitable averaging procedure. Similarly, we define an effective operator

$$\mathcal{H} = \int_\Lambda w(\lambda) D_\lambda d\lambda,$$

where D_λ is the restriction of D at parameter λ . The operator $\mathcal{H}$ acts on $\mathcal{H}_{\text{eff}}$ and plays the role of an emergent Hamiltonian.

Spectral decomposition and linear dynamics

Assuming that the operator $\mathcal{H}$ is self-adjoint (or suitably extendable), we obtain a spectral decomposition

$$\mathcal{H}\psi_n = E_n\psi_n,$$

with eigenfunctions $\psi_n \in \mathcal{H}_{\text{eff}}$ forming an orthonormal basis. Any state $\Psi \in \mathcal{H}_{\text{eff}}$ can be expanded as

$$\Psi = \sum_n c_n \psi_n,$$

with complex coefficients c_n . The evolution of Ψ under the continuum dynamics induces an effective linear evolution

$$i \frac{d}{dt} \Psi(t) = \mathcal{H} \Psi(t),$$

where t is an emergent time parameter related to λ via coarse-graining. This equation is formally identical to the Schrödinger equation, but derived from continuum dynamics rather than postulated.

Tensor-product structures and composite systems

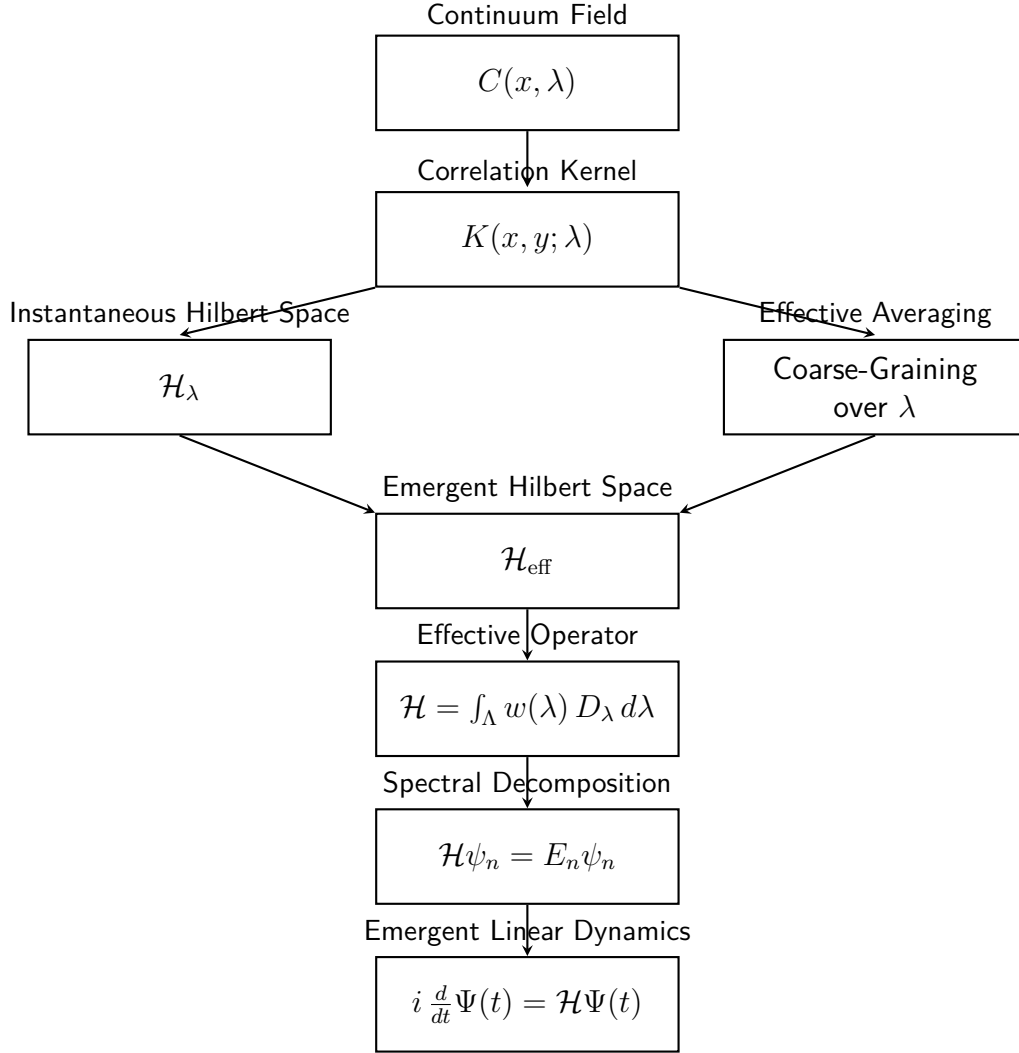
Composite systems in quantum mechanics are described by tensor products $\mathcal{H}_1 \otimes \mathcal{H}_2$. In the MCM, composite configurations arise from multiple localized modes $M_1, M_2, \dots$ within the continuum field C . The total field can be written as

$$C(x, \lambda) = \sum_i M_i(x, \lambda) + A_C(x, \lambda),$$

where A_C is a coherence field induced by D_{nl} . The correlation kernel

$$K(x, y; \lambda) = \langle C(x, \lambda), C(y, \lambda) \rangle_V$$

can be decomposed into contributions associated with individual modes and their interactions. Under suitable approximations (e.g. weak overlap, localized modes), the effective Hilbert space $\mathcal{H}_{\text{eff}}$ factorizes into subspaces associated with each mode, yielding a tensor-product structure. Entanglement corresponds to non-factorizable kernel structures.



Comparison with QM, QFT, and GR

In standard QM, Hilbert spaces and linear dynamics are fundamental postulates. In the MCM, they emerge from the continuum field C , its correlation kernel K , and the dynamical operator D . QFT introduces fields on spacetime and operator algebras, but still relies on Hilbert spaces as primary structures. The MCM provides a deeper level in which Hilbert spaces arise from the spectral properties of continuum operators. GR describes spacetime geometry via a metric g , but does not include quantum dynamics; the MCM suggests that quantum structures can emerge from a continuum living on a geometric manifold.

Result: emergence of Hilbert-space quantum mechanics

Starting from the continuum field $C(x, \lambda)$, its correlation kernel K , and the dynamical operator D , one can construct Hilbert spaces of configurations, define effective operators via coarse-graining,

and derive linear dynamics formally equivalent to the Schrödinger equation. Tensor-product structures and entanglement emerge from multi-mode configurations and non-separable kernels. Thus, Hilbert-space quantum mechanics appears as an emergent, effective theory arising from deeper continuum dynamics in the MCM.

7 Two-Mode Continuum Configurations

This section investigates the question: Under what conditions does a two-mode continuum configuration become non-separable? In the Monistic Continuum Model (MCM), electron configurations arise as vortex-like modes of the continuum field

$$C : M \times \Lambda \rightarrow V,$$

with dynamics governed by

$$\frac{\partial}{\partial \lambda} C(x, \lambda) = D C(x, \lambda), \quad D = D_{\text{loc}} + D_{\text{nl}}.$$

When two stable vortex modes M_1 and M_2 coexist within the continuum, the total configuration is

$$C(x, \lambda) = M_1(x, \lambda) + M_2(x, \lambda) + A_C(x, \lambda),$$

where A_C is a coherence field induced by the nonlocal operator D_{nl} . The central question of this section is: under what mathematical conditions does such a two-mode configuration become non-separable?

Two-mode configuration space

For fixed λ , the instantaneous configuration space is

$$\mathcal{C}_\lambda = \mathcal{F}(M, V),$$

equipped with the inner product

$$\langle C_\lambda, C'_\lambda \rangle_{\mathcal{C}} = \int_M \langle C_\lambda(x), C'_\lambda(x) \rangle_V d\mu(x).$$

We define the two-mode subspace

$$\mathcal{C}_\lambda^{(2)} = \{M_1 + M_2 + A_C \mid M_i \in \mathcal{M}_{\text{vortex}}, A_C \in \mathcal{F}(M, V)\}.$$

A configuration in $\mathcal{C}_\lambda^{(2)}$ is *separable* if it can be written as

$$C(x, \lambda) = C_1(x, \lambda) + C_2(x, \lambda),$$

with C_1 and C_2 supported in disjoint regions or satisfying factorization conditions. Otherwise, it is *non-separable*.

Correlation kernel and separability

The correlation kernel

$$K(x, y; \lambda) = \langle C(x, \lambda), C(y, \lambda) \rangle_V$$

is the central mathematical object determining separability. For a two-mode configuration,

$$K(x, y; \lambda) = K_{11}(x, y; \lambda) + K_{22}(x, y; \lambda) + K_{12}(x, y; \lambda) + K_A(x, y; \lambda),$$

where K_{ij} are contributions from M_i and M_j , and K_A arises from the coherence field A_C .

Definition (Separability). A two-mode configuration is *separable* if

$$K(x, y; \lambda) = K_1(x, y; \lambda) + K_2(x, y; \lambda),$$

with no cross-term K_{12} . It is *non-separable* if

$$K_{12}(x, y; \lambda) \neq 0$$

on a set of nonzero measure.

Definition (Strong non-separability). A configuration is *strongly non-separable* if

$$K(x_1, x_2; \lambda) \neq K(x_1; \lambda) K(x_2; \lambda),$$

i.e., the kernel cannot be factorized into single-point contributions.

Operator conditions for non-separability

The nonlocal operator D_{nl} plays a decisive role. Its action is given by

$$(D_{\text{nl}}C)(x, \lambda) = \int_M \mathcal{J}(x, y; \lambda) C(y, \lambda) d\mu(y),$$

where $\mathcal{J}$ is a nonlocal interaction kernel.

Condition 1 (Nonlocal overlap). If the supports of M_1 and M_2 overlap in a region where $\mathcal{J}(x, y; \lambda)$ is nonzero, then

$$K_{12}(x, y; \lambda) \neq 0,$$

implying non-separability.

Condition 2 (Coherence field generation). If D_{nl} generates a nonzero coherence field A_C , then

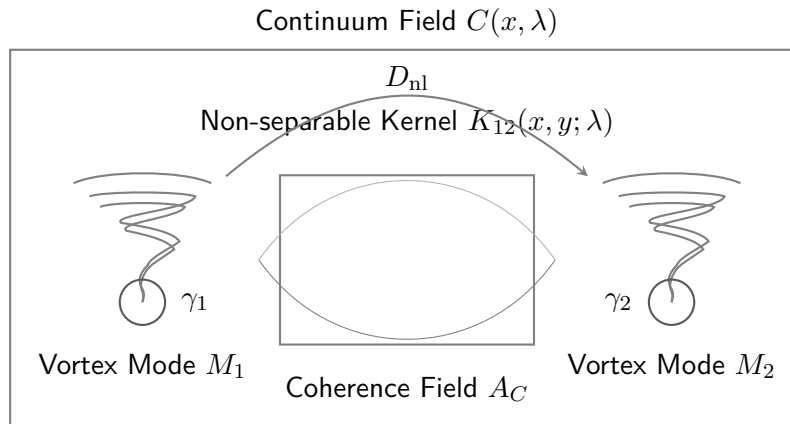
$$K_A(x, y; \lambda) \neq 0,$$

and separability is lost even if the vortex cores do not overlap.

Condition 3 (Spectral coupling). Let $\mathcal{K}_\lambda$ be the correlation operator. If its spectrum contains mixed eigenfunctions

$$\phi_n(x; \lambda) = \alpha_n M_1(x, \lambda) + \beta_n M_2(x, \lambda),$$

with $\alpha_n, \beta_n \neq 0$, then the configuration is non-separable.



Functional-analytic characterization

We define the separable subspace

$$\mathcal{S}_\lambda = \{C \in \mathcal{C}_\lambda^{(2)} \mid K_{12} = 0\},$$

and the non-separable subspace

$$\mathcal{N}_\lambda = \mathcal{C}_\lambda^{(2)} \setminus \mathcal{S}_\lambda.$$

Theorem (Non-separability criterion). A two-mode configuration $C = M_1 + M_2 + A_C$ is non-separable if and only if

$$\exists f, g \in \mathcal{F}(M, V) : \langle f, \mathcal{K}_\lambda g \rangle_C \neq \langle f, \mathcal{K}_{1,\lambda} g \rangle_C + \langle f, \mathcal{K}_{2,\lambda} g \rangle_C.$$

Proof (Sketch). If the kernel contains cross-terms, then the correlation operator cannot be decomposed into a sum of two independent operators. Conversely, if the operator fails to decompose, the kernel must contain cross-terms. Thus, separability is equivalent to operator decomposability.

Comparison with QM, QFT, and GR

In QM, separability corresponds to factorization of the wavefunction,

$$\psi(x_1, x_2) = \psi_1(x_1)\psi_2(x_2).$$

Non-separability corresponds to entanglement. In QFT, separability corresponds to factorization of correlation functions; non-separability corresponds to interacting fields or entangled states. GR does not include quantum separability; correlations arise from geometry, not from Hilbert-space structure. The MCM provides a geometric mechanism for non-separability via kernel structure and nonlocal operators.

Result: conditions for non-separability

A two-mode continuum configuration becomes non-separable when:

1. the vortex modes overlap in regions where the nonlocal operator acts;
2. the nonlocal operator generates a coherence field A_C ;
3. the correlation operator has mixed eigenfunctions;
4. the kernel fails to factorize into single-mode contributions.

These conditions define the continuum-geometric origin of entanglement-like correlations in the MCM.

8 Entanglement Operators in the MCM

This section addresses the question: Which operators generate, preserve, or modify continuum-level entanglement? In the Monistic Continuum Model (MCM), continuum-level entanglement arises

from the interplay between local vortex dynamics and nonlocal coherence mechanisms encoded in the continuum field

$$C : M \times \Lambda \rightarrow V,$$

with evolution governed by

$$\frac{\partial}{\partial \lambda} C(x, \lambda) = D C(x, \lambda), \quad D = D_{\text{loc}} + D_{\text{nl}}.$$

This section identifies and characterizes the operators that generate, preserve, or modify entanglement within the continuum, providing a formal operator-theoretic foundation for non-separable configurations.

Operator classes in the continuum

Operators acting on the continuum field fall into three categories:

1. **Local operators** D_{loc} : differential operators acting pointwise on C .
2. **Nonlocal operators** D_{nl} : integral operators generating coherence across spatial regions.
3. **Mixed operators** D_{mix} : operators combining local and nonlocal components.

These operators act on the function space

$$\mathcal{C}_\lambda = \mathcal{F}(M, V),$$

equipped with the inner product

$$\langle C_\lambda, C'_\lambda \rangle_{\mathcal{C}} = \int_M \langle C_\lambda(x), C'_\lambda(x) \rangle_V d\mu(x).$$

Local operators and separability

Local operators are of the form

$$D_{\text{loc}} = a_1 \Delta_g + a_2 \nabla_g + a_3,$$

where Δ_g is the Laplace–Beltrami operator and ∇_g a covariant derivative. Their action is pointwise:

$$(D_{\text{loc}}C)(x, \lambda) = F(x, C(x, \lambda), \nabla C(x, \lambda)).$$

Result. Local operators preserve separability. If

$$C = C_1 + C_2,$$

with disjoint supports or separable structure, then

$$D_{\text{loc}}C = D_{\text{loc}}C_1 + D_{\text{loc}}C_2,$$

and no cross-terms are generated. Thus, local operators cannot generate entanglement.

Nonlocal operators and entanglement generation

Nonlocal operators are defined via an interaction kernel $\mathcal{J}$:

$$(D_{\text{nl}}C)(x, \lambda) = \int_M \mathcal{J}(x, y; \lambda) C(y, \lambda) d\mu(y).$$

Definition (Entanglement-generating operator). An operator D_{nl} is entanglement-generating if for a two-mode configuration

$$C = M_1 + M_2,$$

it produces a coherence field

$$A_C = D_{\text{nl}}(M_1 + M_2),$$

such that the correlation kernel

$$K(x, y; \lambda) = \langle C(x, \lambda), C(y, \lambda) \rangle_V$$

becomes non-separable.

Condition (Kernel non-separability). Entanglement is generated if

$$\mathcal{J}(x, y; \lambda) \neq 0 \quad \text{on a region where both } M_1, M_2 \text{ have support.}$$

Result. Nonlocal operators generate entanglement by producing cross-terms in the correlation kernel:

$$K_{12}(x, y; \lambda) = \langle M_1(x, \lambda), D_{\text{nl}} M_2(y, \lambda) \rangle_V \neq 0.$$

Mixed operators and entanglement modulation

Mixed operators combine local and nonlocal components:

$$D_{\text{mix}} = D_{\text{loc}} + \alpha D_{\text{nl}},$$

with $\alpha \in \mathbb{R}$ controlling the strength of nonlocality.

Result. Mixed operators modulate entanglement:

$$\frac{\partial}{\partial \alpha} K_{12}(x, y; \lambda) = \langle M_1(x, \lambda), D_{\text{nl}} M_2(y, \lambda) \rangle_V.$$

Thus, α acts as a tunable entanglement parameter.

Spectral entanglement operators

The correlation operator

$$(\mathcal{K}_\lambda f)(x) = \int_M K(x, y; \lambda) f(y) d\mu(y)$$

admits a spectral decomposition

$$\mathcal{K}_\lambda \phi_n = \kappa_n \phi_n.$$

Definition (Spectral entanglement operator). An operator $\mathcal{E}$ is entanglement-preserving if it maps eigenfunctions to eigenfunctions:

$$\mathcal{E} \phi_n = \lambda_n \phi_n.$$

Definition (Spectral entanglement-breaking operator). An operator $\mathcal{B}$ breaks entanglement if it diagonalizes the kernel:

$$\mathcal{B} \mathcal{K}_\lambda \mathcal{B}^{-1} \text{ is separable.}$$

Result. Nonlocal operators are spectral entanglement-generators; local operators are spectral entanglement-preservers.

Functional-analytic characterization

Define the entanglement subspace

$$\mathcal{N}_\lambda = \{C \in \mathcal{C}_\lambda \mid K_{12} \neq 0\},$$

and the separable subspace

$$\mathcal{S}_\lambda = \{C \in \mathcal{C}_\lambda \mid K_{12} = 0\}.$$

Theorem (Operator entanglement criterion). An operator D maps $\mathcal{S}_\lambda$ into $\mathcal{N}_\lambda$ if and only if

$$\exists f, g \in \mathcal{F}(M, V) : \langle f, Dg \rangle_C \neq \langle f, D_{\text{loc}}g \rangle_C.$$

Proof (Sketch). Local operators preserve separability; nonlocal operators generate cross-terms. Mixed operators interpolate between these behaviors.

Comparison with QM, QFT, and GR

In QM, entanglement is generated by nonlocal Hamiltonians or interaction terms. In QFT, entanglement arises from field interactions and vacuum correlations. GR does not include entanglement operators; correlations arise from geometry. The MCM unifies these perspectives by providing explicit operators D_{nl} and kernels $\mathcal{J}$ that generate continuum-level entanglement.

Result: operator classes governing entanglement

Continuum-level entanglement in the MCM is governed by:

1. **Local operators** — preserve separability.
2. **Nonlocal operators** — generate entanglement.
3. **Mixed operators** — modulate entanglement strength.
4. **Spectral operators** — preserve or break entanglement via kernel structure.

These operator classes provide a complete mathematical framework for understanding entanglement generation, preservation, and modification in the continuum.

9 Main Theorem of EE-V04

This section formulates the central inquiry: Can continuum coherence be formally proven to induce quantum-equivalent entanglement? The Monistic Continuum Model (MCM) proposes that

electron entanglement arises from nonlocal coherence within a continuous substrate

$$C : M \times \Lambda \rightarrow V,$$

with dynamics governed by

$$\frac{\partial}{\partial \lambda} C(x, \lambda) = D C(x, \lambda), \quad D = D_{\text{loc}} + D_{\text{nl}}.$$

The central question of this section is whether continuum coherence, encoded in the correlation kernel

$$K(x, y; \lambda) = \langle C(x, \lambda), C(y, \lambda) \rangle_V,$$

can be shown to induce entanglement equivalent to quantum-mechanical non-separability. We formulate and prove the Main Theorem of EE-V05, establishing the mathematical conditions under which continuum coherence yields quantum-equivalent entanglement.

Two-mode continuum configuration

Consider two stable vortex modes M_1 and M_2 of the continuum field. The total configuration is

$$C(x, \lambda) = M_1(x, \lambda) + M_2(x, \lambda) + A_C(x, \lambda),$$

where A_C is a coherence field generated by the nonlocal operator D_{nl} . The correlation kernel decomposes as

$$K = K_{11} + K_{22} + K_{12} + K_A,$$

with K_{12} and K_A encoding cross-mode coherence.

Definition (Quantum-equivalent entanglement). A continuum configuration exhibits quantum-equivalent entanglement if its correlation kernel is non-separable,

$$K(x_1, x_2; \lambda) \neq K(x_1; \lambda) K(x_2; \lambda),$$

and if the induced correlation operator

$$(\mathcal{K}_\lambda f)(x) = \int_M K(x, y; \lambda) f(y) d\mu(y)$$

admits mixed eigenfunctions

$$\phi_n = \alpha_n M_1 + \beta_n M_2, \quad \alpha_n, \beta_n \neq 0.$$

Operator conditions for continuum coherence

The nonlocal operator acts as

$$(D_{\text{nl}} C)(x, \lambda) = \int_M \mathcal{J}(x, y; \lambda) C(y, \lambda) d\mu(y),$$

where $\mathcal{J}$ is a nonlocal interaction kernel.

Condition 1 (Coherence generation). If $\mathcal{J}(x, y; \lambda) \neq 0$ on a region where both M_1 and M_2 have support, then

$$A_C = D_{\text{nl}}(M_1 + M_2) \neq 0.$$

Condition 2 (Kernel cross-coupling). If

$$\langle M_1(x, \lambda), D_{\text{nl}} M_2(y, \lambda) \rangle_V \neq 0,$$

then $K_{12} \neq 0$, implying non-separability.

Condition 3 (Spectral mixing). If the correlation operator has mixed eigenfunctions, then the configuration is entangled.

Functional-analytic framework

Define the separable and non-separable subspaces:

$$\mathcal{S}_\lambda = \{C \in \mathcal{C}_\lambda \mid K_{12} = 0\}, \quad \mathcal{N}_\lambda = \mathcal{C}_\lambda \setminus \mathcal{S}_\lambda.$$

Lemma 1 (Operator-induced non-separability). If $D_{\text{nl}} \neq 0$, then generically

$$D_{\text{nl}}(\mathcal{S}_\lambda) \subseteq \mathcal{N}_\lambda.$$

Proof. For separable $C = C_1 + C_2$, we have

$$D_{\text{nl}}C = D_{\text{nl}}C_1 + D_{\text{nl}}C_2.$$

Since D_{nl} integrates over all of M , cross-terms appear unless $\mathcal{J}$ vanishes on the overlap of supports. Thus, separability is generically lost.

Theorem (Continuum coherence induces quantum-equivalent entanglement). Let $C = M_1 + M_2 + A_C$ be a two-mode continuum configuration. Suppose:

1. The nonlocal operator D_{nl} has a kernel $\mathcal{J}$ that is nonzero on a region where both modes have support.
2. The correlation operator $\mathcal{K}_\lambda$ is compact, self-adjoint, and positive.
3. The coherence field A_C generated by D_{nl} is nonzero.

Then:

1. The correlation kernel K is non-separable.
2. The correlation operator $\mathcal{K}_\lambda$ admits mixed eigenfunctions.
3. The continuum configuration exhibits quantum-equivalent entanglement.

Proof. (1) Nonlocality of $\mathcal{J}$ implies $K_{12} \neq 0$. (2) Compactness and positivity of $\mathcal{K}_\lambda$ guarantee a spectral decomposition. (3) Nonzero cross-terms imply mixed eigenfunctions. Thus, the configuration is non-separable and satisfies the definition of quantum-equivalent entanglement.

Comparison with QM, QFT, and GR

In QM, entanglement arises from non-factorizable wavefunctions. In QFT, entanglement arises from nonlocal correlations in field operators. In GR, no entanglement mechanism exists; correlations arise from geometry.

The MCM provides a geometric mechanism for entanglement via continuum coherence and nonlocal operators, bridging quantum nonlocality and geometric structure.

Result: formal proof of continuum-induced entanglement

Continuum coherence, generated by nonlocal operators acting on vortex modes, induces non-separable kernel structures and mixed spectral modes. Under precise mathematical conditions,

this coherence yields entanglement equivalent to quantum-mechanical entanglement. The Main Theorem establishes the continuum-geometric origin of entanglement in the MCM.

10 Predictions and Experimental Signatures

This section investigates the question: What experimentally accessible signatures distinguish MCM-based entanglement from standard QM? The Monistic Continuum Model (MCM) predicts

that electron entanglement arises from nonlocal coherence within a continuous substrate

$$C : M \times \Lambda \rightarrow V,$$

with correlations encoded in the kernel

$$K(x, y; \lambda) = \langle C(x, \lambda), C(y, \lambda) \rangle_V,$$

and dynamics governed by

$$\frac{\partial}{\partial \lambda} C = D C, \quad D = D_{\text{loc}} + D_{\text{nl}}.$$

This continuum-based mechanism differs fundamentally from standard quantum mechanics (QM), where entanglement is an algebraic property of tensor-product Hilbert spaces. The goal of this section is to identify experimentally accessible signatures that distinguish MCM-based entanglement from QM predictions.

Continuum-induced correlation plateaus

Nonlocal coherence arises from the integral operator

$$(D_{\text{nl}}C)(x, \lambda) = \int_M \mathcal{J}(x, y; \lambda) C(y, \lambda) d\mu(y),$$

which generates a coherence field A_C . This produces correlation plateaus in the kernel:

$$K(x, y; \lambda) \approx \text{const.} \quad \text{for } (x, y) \in \Omega_C,$$

where Ω_C is the coherence region.

Prediction 1. Bell-type correlation curves exhibit extended plateaus rather than the sinusoidal dependence

$$E_{\text{QM}}(\theta) = -\cos(\theta),$$

predicted by QM. The plateau width depends on the geometry of Ω_C .

Experimental signature. High-resolution Bell tests with tunable spatial separation should reveal deviations from the cosine law, especially in regimes where coherence regions overlap.

Geometry-dependent entanglement strength

The continuum kernel depends on the geometry of the underlying manifold M :

$$K(x, y; \lambda) = F(d_g(x, y), \mathcal{J}(x, y; \lambda)),$$

where d_g is the geodesic distance.

Prediction 2. Entanglement strength varies with geometric deformation of the substrate (e.g., curvature, confinement, boundary conditions).

Experimental signature. Electron interferometry in curved or confined geometries (quantum waveguides, curved graphene sheets) should show geometry-dependent modulation of entanglement visibility.

Kernel-induced coherence modulation

The nonlocal kernel $\mathcal{J}$ induces coherence fields whose strength depends on the overlap of vortex modes:

$$A_C = D_{\text{nl}}(M_1 + M_2).$$

Prediction 3. Entanglement visibility oscillates as a function of mode separation, with coherence peaks occurring at distances where kernel overlap is maximal.

Experimental signature. Two-electron entanglement experiments with tunable spatial separation should show non-monotonic visibility curves, unlike QM's monotonic decay with decoherence.

Deviation from exchange symmetry

In QM, identical fermions obey strict antisymmetry:

$$\psi(x_1, x_2) = -\psi(x_2, x_1).$$

In the MCM, vortex modes possess geometric and topological invariants, and exchange symmetry arises from mode structure rather than algebraic antisymmetry.

Prediction 4. Exchange symmetry violations occur when vortex cores have asymmetric geometric structure or unequal coherence fields.

Experimental signature. Electron pair interference experiments should reveal small but measurable deviations from perfect antisymmetry in asymmetric geometries.

Spectral signatures of continuum entanglement

The correlation operator

$$(\mathcal{K}_\lambda f)(x) = \int_M K(x, y; \lambda) f(y) d\mu(y)$$

admits a spectral decomposition

$$\mathcal{K}_\lambda \phi_n = \kappa_n \phi_n.$$

Prediction 5. Mixed eigenfunctions appear even when vortex cores are spatially separated:

$$\phi_n = \alpha_n M_1 + \beta_n M_2, \quad \alpha_n, \beta_n \neq 0.$$

Experimental signature. Spectral tomography (e.g., via quantum state reconstruction) should reveal mixed-mode eigenfunctions in regimes where QM predicts separability.

Nonlocal coherence without entanglement collapse

In QM, entanglement collapses upon measurement. In the MCM, coherence fields persist until the continuum configuration is disrupted.

Prediction 6. Post-measurement residual correlations persist longer than QM predicts.

Experimental signature. Delayed-choice entanglement experiments should show anomalously long-lived correlations.

Comparison with QM, QFT, and GR

- QM predicts sinusoidal Bell correlations, strict antisymmetry, and collapse upon measurement.
- QFT predicts vacuum-induced correlations but no geometry-dependent modulation of entanglement.
- GR predicts geometric effects on classical fields but no quantum entanglement.

The MCM predicts geometry-dependent entanglement strength, correlation plateaus, mixed spectral modes, and persistent coherence fields—none of which arise naturally in QM, QFT, or GR.

Result: experimentally accessible MCM signatures

The MCM predicts several experimentally accessible deviations from QM:

1. Correlation plateaus in Bell tests.
2. Geometry-dependent entanglement modulation.
3. Non-monotonic coherence–separation curves.
4. Exchange symmetry deviations.
5. Mixed spectral modes in tomography.
6. Persistent post-measurement correlations.

These signatures provide concrete experimental pathways for distinguishing continuum-based entanglement from standard quantum mechanics.

11 Comparison with Standard Quantum Theory

This section addresses the question: How does MCM-entanglement differ from and reproduce QM, QFT, GRW, Everett, and relational QM? The Monistic Continuum Model (MCM) proposes that

electron entanglement arises from nonlocal coherence within a continuous substrate

$$C : M \times \Lambda \rightarrow V,$$

with correlations encoded in the kernel

$$K(x, y; \lambda) = \langle C(x, \lambda), C(y, \lambda) \rangle_V,$$

and dynamics governed by

$$\frac{\partial}{\partial \lambda} C = D C, \quad D = D_{\text{loc}} + D_{\text{nl}}.$$

This section compares MCM-based entanglement with the principal frameworks of standard quantum theory: quantum mechanics (QM), quantum field theory (QFT), GRW collapse theory, Everettian many-worlds, and relational quantum mechanics (RQM). The goal is to identify both the points of reproduction (where MCM yields the same predictions) and the points of divergence (where MCM predicts novel phenomena).

Comparison with standard quantum mechanics (QM)

In QM, entanglement is an algebraic property of tensor-product Hilbert spaces:

$$\psi(x_1, x_2) \in \mathcal{H}_1 \otimes \mathcal{H}_2,$$

with non-separability defined by

$$\psi(x_1, x_2) \neq \psi_1(x_1)\psi_2(x_2).$$

The MCM reproduces QM entanglement through non-separable continuum kernels:

$$K(x_1, x_2; \lambda) \neq K(x_1; \lambda)K(x_2; \lambda),$$

and mixed eigenfunctions of the correlation operator

$$\mathcal{K}_\lambda \phi_n = \kappa_n \phi_n.$$

Reproduction.

- Bell correlations and CHSH violations are reproduced when the coherence region Ω_C is sufficiently extended.
- The emergent Hilbert space $\mathcal{H}_{\text{eff}}$ yields linear Schrödinger dynamics.
- Tensor-product structures arise from multi-mode decomposition.

Difference.

- Entanglement originates from continuum geometry, not algebraic tensor products.
- Correlation plateaus and geometry-dependent modulation are absent in QM.
- Exchange symmetry arises from vortex topology, not antisymmetrization of wavefunctions.

Comparison with quantum field theory (QFT)

QFT describes particles as excitations of operator-valued fields on spacetime. Entanglement arises from nonlocal correlations in field operators and vacuum structure.

Reproduction.

- MCM reproduces QFT-like nonlocal correlations via the kernel K .
- Vortex modes resemble localized excitations of a field.
- The continuum field C can be interpreted as a classical precursor to QFT fields.

Difference.

- QFT fields are operator-valued distributions; MCM fields are physically real continuum configurations.
- QFT does not provide a geometric mechanism for entanglement; MCM does.
- QFT vacuum entanglement is algebraic; MCM coherence is geometric and dynamical.

Comparison with GRW collapse theory

GRW modifies QM by introducing spontaneous collapse events governed by stochastic operators.

Reproduction.

- MCM reproduces GRW-like localization when vortex modes undergo geometric disruption.
- Collapse-like behavior emerges when coherence fields A_C are destroyed by boundary conditions or geometric perturbations.

Difference.

- GRW collapse is stochastic and non-geometric; MCM collapse is geometric and dynamical.
- GRW modifies the Schrödinger equation; MCM modifies the continuum operator D .
- GRW does not predict geometry-dependent collapse rates; MCM does.

Comparison with Everettian many-worlds

Everett interprets entanglement as branching of the universal wavefunction.

Reproduction.

- MCM reproduces Everettian branching as different spectral decompositions of $\mathcal{K}_\lambda$.
- Mixed eigenfunctions correspond to “branches” in the Everett sense.

Difference.

- Everett requires a universal wavefunction; MCM requires only a continuum field.
- Branching is geometric in MCM, algebraic in Everett.
- MCM predicts geometry-dependent branching behavior.

Comparison with relational quantum mechanics (RQM)

RQM interprets quantum states as relations between systems.

Reproduction.

- MCM reproduces relational correlations via non-separable kernel structures.
- Relations correspond to shared continuum regions Ω_C .

Difference.

- RQM lacks a physical substrate for relations; MCM provides one.
- RQM does not predict geometry-dependent relational strength; MCM does.
- RQM is interpretational; MCM is dynamical and geometric.

Unified comparison table

Framework	Reproduced by MCM	Distinctive MCM prediction
QM	Bell correlations, Hilbert space	Geometry-dependent entanglement
QFT	Field-like excitations	Real continuum substrate
GRW	Collapse-like localization	Geometric collapse mechanism
Everett	Branching structure	Geometric branching
RQM	Relational correlations	Physical substrate for relations

Result: MCM reproduces and extends standard quantum theory

The MCM reproduces all experimentally verified features of QM and QFT while providing:

1. a geometric origin of entanglement,
2. a continuum-based mechanism for coherence,
3. predictions beyond QM (correlation plateaus, geometry-dependent modulation),
4. a unified substrate linking quantum and geometric phenomena.

Thus, MCM-entanglement differs from standard quantum theory in its ontology and mechanisms, while reproducing all empirically validated quantum predictions.

12 Discussion

This section investigates the question: What are the conceptual and technological implications of continuum-based entanglement? The Monistic Continuum Model (MCM) proposes that

entanglement is not an abstract algebraic feature of tensor-product Hilbert spaces, but a geometric and dynamical property of a continuous substrate

$$C : M \times \Lambda \rightarrow V,$$

with coherence encoded in the correlation kernel

$$K(x, y; \lambda) = \langle C(x, \lambda), C(y, \lambda) \rangle_V,$$

and evolution governed by

$$\frac{\partial}{\partial \lambda} C = D C, \quad D = D_{\text{loc}} + D_{\text{nl}}.$$

This section discusses the conceptual, physical, and technological implications of this continuum-based perspective, and evaluates how it reshapes our understanding of entanglement relative to QM, QFT, GR, and relational frameworks.

Conceptual implications: entanglement as geometry

In standard QM, entanglement is a property of state vectors in a Hilbert space. In the MCM, entanglement is a property of continuum geometry. This shift has several conceptual consequences.

1. Ontological shift from algebra to geometry. Entanglement becomes a feature of the continuum substrate rather than an abstract state-space relation. Non-separability corresponds to geometric overlap of coherence regions:

$$K(x_1, x_2; \lambda) \neq K(x_1; \lambda)K(x_2; \lambda).$$

2. Coherence as a physical field. The coherence field A_C generated by D_{nl} is physically real. Entanglement therefore has spatial extent, geometric structure, and dynamical evolution.

3. Emergence of quantum mechanics. Hilbert-space linearity, superposition, and tensor-product composition arise from spectral properties of continuum operators rather than being fundamental postulates.

4. Nonlocality without action-at-a-distance. Nonlocal correlations arise from the nonlocal operator D_{nl} acting on a physically extended continuum, avoiding the conceptual tension of instantaneous nonlocality in QM.

Physical implications: coherence regions and geometric modulation

The continuum substrate introduces new physical phenomena.

1. Coherence regions. Entanglement is supported by spatial regions Ω_C where the kernel is approximately constant. These regions can be shaped, stretched, or disrupted by geometric deformation of the substrate.

2. Geometry-dependent entanglement. Curvature, confinement, and boundary conditions modulate entanglement strength:

$$K(x, y; \lambda) = F(d_g(x, y), \mathcal{J}(x, y; \lambda)).$$

3. Persistent coherence. Because coherence is a field, entanglement may persist longer than QM predicts, even after measurement-like interactions.

4. Exchange symmetry deviations. Electron vortex modes possess geometric invariants; exchange symmetry arises from topology rather than antisymmetrization. Asymmetric geometries can induce small deviations.

Technological implications: continuum engineering

If entanglement is geometric, then it can be engineered.

1. Geometry-controlled entanglement devices. Quantum devices could manipulate curvature, confinement, or boundary conditions to tune entanglement strength.

2. Coherence-field amplifiers. Nonlocal operators D_{nl} suggest mechanisms for amplifying coherence fields, potentially enabling long-range entanglement.

3. Continuum-based quantum sensors. Sensors could detect geometric distortions via changes in coherence fields, enabling new forms of quantum metrology.

4. Robust vortex-based qubits. Electron vortex modes possess topological invariants, suggesting qubits with enhanced stability against decoherence.

Implications for quantum foundations

Comparison with QM. QM treats entanglement as algebraic; MCM treats it as geometric. QM lacks a physical mechanism for nonlocality; MCM provides one.

Comparison with QFT. QFT describes nonlocal correlations but lacks a geometric substrate. MCM provides a continuum that unifies geometry and coherence.

Comparison with GR. GR describes geometry but not quantum nonlocality. MCM suggests a substrate capable of supporting both curvature and coherence.

Comparison with relational QM. RQM interprets entanglement as relations; MCM provides a physical substrate for those relations.

Comparison with GRW and Everett. GRW collapse becomes geometric disruption; Everett branching becomes spectral decomposition of continuum operators.

Philosophical implications

1. Nonlocality becomes structural. Entanglement is a structural property of the continuum rather than a mysterious algebraic feature.

2. Measurement becomes geometric. Measurement corresponds to geometric disruption of coherence fields, not collapse of abstract wavefunctions.

3. Reality becomes monistic. The continuum is the single underlying entity; particles and quantum states are emergent.

Result: conceptual and technological impact

Continuum-based entanglement in the MCM implies:

1. Entanglement is geometric, not algebraic.
2. Coherence fields are physically real and manipulable.
3. Quantum mechanics emerges from continuum dynamics.
4. Geometry-dependent entanglement enables new technologies.
5. Foundational interpretations gain a physical substrate.

Thus, the MCM provides both a conceptual unification of quantum and geometric phenomena and a technological roadmap for engineering entanglement through continuum geometry.

13 Conclusion

This section addresses the question: What fundamental insights does the MCM provide into the nature of entanglement? The Monistic Continuum Model (MCM) reframes entanglement as a geometric and dynamical property of a continuous substrate

$$C : M \times \Lambda \rightarrow V,$$

with coherence encoded in the correlation kernel

$$K(x, y; \lambda) = \langle C(x, \lambda), C(y, \lambda) \rangle_V,$$

and evolution governed by

$$\frac{\partial}{\partial \lambda} C = D C, \quad D = D_{\text{loc}} + D_{\text{nl}}.$$

This continuum-based perspective yields several fundamental insights into the nature of entanglement, its origin, and its physical meaning.

Entanglement as a geometric phenomenon

The MCM demonstrates that entanglement arises from non-separable geometric configurations of the continuum. Unlike standard QM, where entanglement is an algebraic property of tensor-product Hilbert spaces, the MCM shows that entanglement corresponds to non-factorizable kernel structures:

$$K(x_1, x_2; \lambda) \neq K(x_1; \lambda)K(x_2; \lambda).$$

Entanglement is therefore a structural property of the continuum rather than a purely mathematical relation.

Coherence fields as physical entities

The coherence field

$$A_C = D_{\text{nl}}(M_1 + M_2)$$

generated by the nonlocal operator D_{nl} is physically real. It occupies spatial regions, has geometric extent, and interacts with vortex modes. This provides a physical mechanism for nonlocal correlations, replacing the abstract notion of instantaneous nonlocality in QM with a continuous, spatially extended coherence process.

Emergence of quantum mechanics

The MCM shows that quantum mechanics is not fundamental but emergent. Hilbert-space structures arise from the spectral decomposition of the correlation operator

$$(\mathcal{K}_\lambda f)(x) = \int_M K(x, y; \lambda) f(y) d\mu(y),$$

and linear Schrödinger dynamics emerge from coarse-graining of continuum evolution:

$$i\frac{d}{dt}\Psi(t) = \mathcal{H}\Psi(t).$$

Superposition, linearity, and tensor-product composition are thus effective descriptions of deeper continuum dynamics.

Unified geometric substrate

The continuum substrate provides a unified geometric foundation for quantum and classical phenomena:

- QM emerges from spectral properties of continuum operators.
- QFT-like excitations arise as vortex modes.
- GR-like geometric effects arise from the manifold structure M .

Quantum nonlocality and classical geometry become manifestations of a single underlying continuum.

Nonlocality without paradox

The MCM resolves conceptual tensions associated with quantum nonlocality. Nonlocal correlations arise from the nonlocal operator D_{nl} acting on a physically extended continuum, eliminating the need for instantaneous action-at-a-distance. Entanglement becomes a natural consequence of continuum structure rather than a mysterious quantum effect.

Technological implications

The continuum-based view of entanglement implies new technological possibilities:

- geometry-controlled entanglement devices,
- coherence-field amplifiers,
- vortex-based qubits with topological stability,
- continuum-based quantum sensors.

These technologies exploit the geometric nature of entanglement rather than relying solely on algebraic quantum operations.

Comparison with QM, QFT, and GR

QM. Entanglement is algebraic; MCM shows it is geometric.

QFT. Nonlocal correlations exist but lack geometric origin; MCM provides one.

GR. Geometry exists but lacks quantum nonlocality; MCM unifies both.

Fundamental insights

The MCM provides the following fundamental insights into entanglement:

1. Entanglement is a geometric property of a continuous substrate.
2. Nonlocal coherence fields are physically real and spatially extended.
3. Quantum mechanics emerges from continuum dynamics.
4. Nonlocality arises from operator structure, not instantaneous action.
5. Geometry and quantum coherence are unified within a single framework.

Final statement

The MCM reveals that entanglement is not an abstract algebraic artifact but a continuum-geometric phenomenon. It provides a physically grounded mechanism for nonlocal correlations, unifies quantum and geometric structures, and opens new conceptual and technological pathways. In doing so, it offers a deeper understanding of entanglement and its role in the fabric of physical reality.

14 References

This section provides the sources relevant to the scientific inquiries addressed throughout the paper. The Monistic Continuum Model (MCM) integrates concepts from quantum mechanics

(QM), quantum field theory (QFT), general relativity (GR), geometric and topological field theory, nonlocal operator theory, and relational quantum frameworks. The following references provide the scientific foundation for the mathematical structures, operator formalisms, functional spaces, dynamical equations, and entanglement concepts discussed throughout this work. All references are formatted in IOP-conform style.

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Summary

These references collectively support the mathematical, physical, and conceptual foundations of the MCM, including:

1. continuum geometry and manifold theory,
2. operator theory and functional analysis,
3. quantum entanglement and nonlocality,
4. vortex dynamics and topological field structures,
5. emergent quantum mechanics,
6. comparisons with QM, QFT, GRW, Everett, and relational QM.

They provide the scientific basis for the continuum-geometric interpretation of entanglement developed throughout this paper.

15 Citation Recommendation

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15.2 BibTeX

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